



CONTINGENCY LESSONS

Mathematics for SylvanSync™

Early Elementary



GRADES K-2

Mathematics for SylvanSync™

Grades K–2

VersaTiles: Using Numbers

Objectives

The student will be able to:

- name a number to 1,000 using place-value models.
- solve problems involving fractional parts.

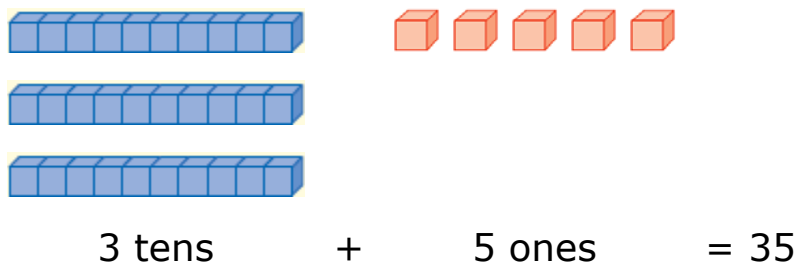
Materials

- *VersaTiles: Using Numbers* book (VT2UN)
- VersaTiles

Background Information

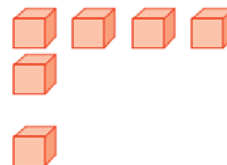
VersaTiles let you learn, practice, and review skills all by yourself. Start by opening the Answer Case. Place the number tiles on the top half of the case with the number side up. The letters A–L will be on the bottom. Answer the questions in your activity book, close the case and flip it over, and check to see if your answers match. VersaTiles activities can help you learn about numbers and problem solving as you practice and strengthen your math skills.

- What number is shown with the models?



Mathematics for SylvanSync™
Grades K–2
VersaTiles: Using Numbers

- What number is shown with the models?



2 hundreds

1 ten

+

6 ones

=

216

- Use the picture to solve the problem.

①	①	①	

Three sections are planted with sunflowers. What fraction of the garden is planted with sunflowers? (Only $\frac{3}{8}$ of the garden is planted with sunflowers.)

What fraction of the garden is NOT planted with sunflowers? (That leaves $\frac{5}{8}$ of the garden not planted with sunflowers.)

Mathematics for SylvanSync™
Grades K–2
VersaTiles: Using Numbers

Activity

- Go to page 6 in *VersaTiles: Using Numbers* to practice naming numbers using place-value models. Flip over the box to see if you did it correctly.
- Go to page 8 in *VersaTiles: Using Numbers* to practice naming numbers using place-value models. Flip over the box to see if you did it correctly.
- Go to page 16 in *VersaTiles: Using Numbers* to practice naming numbers using place-value models. Flip over the box to see if you did it correctly.
- Go to page 30 in *VersaTiles: Using Numbers* to practice solving problems with fractional parts. Flip over the box to see if you did it correctly.

Mathematics for SylvanSync™
Grades K–2
VersaTiles: Number and Shape Patterns

Objectives

The student will be able to:

- identify and continue a number pattern.
- find a function rule.

Materials

- *VersaTiles: Number and Shape Patterns* book (VT2NSP)
- VersaTiles

Background Information

VersaTiles let you learn, practice, and review skills all by yourself. Start by opening the Answer Case. Place the number tiles on the top half of the case with the number side up. The letters A–L will be on the bottom. Answer the questions in your activity book, close the case and flip it over, and check to see if your answers match. VersaTiles activities can help you learn about numbers and problem solving as you practice and strengthen your math skills.

- What is the next number in the pattern?

42, 41, 40, 42, 41, 40, 42, _____

The next number is 41.

- What is the rule for the function machine?

IN		RULE		OUT
4	→	?	→	7

If 4 goes into the function machine and 7 comes out then the rule is + 3.

Mathematics for SylvanSync™
Grades K–2
VersaTiles: Number and Shape Patterns

Activity

- Go to page 10 in *VersaTiles: Number and Shape Patterns* to practice continuing a number pattern by counting by 2 or 3. Flip over the box to see if you did it correctly.
- Go to page 13 in *VersaTiles: Number and Shape Patterns* to practice continuing a number pattern. Flip over the box to see if you did it correctly.
- Go to page 20 in *VersaTiles: Number and Shape Patterns* to practice identifying and continuing a number pattern. Flip over the box to see if you did it correctly.
- Go to page 16 in *VersaTiles: Number and Shape Patterns* to practice finding a function rule. Flip over the box to see if you did it correctly.

Mathematics for SylvanSync™
Grades K–2
VersaTiles: Exploring Data

Objectives

The student will be able to:

- interpret and compare data displayed in graphs.
- identify and list possible combinations.

Materials

- *VersaTiles: Exploring Data* book (VT2ED)
- VersaTiles

Background Information

VersaTiles let you learn, practice, and review skills all by yourself. Start by opening the Answer Case. Place the number tiles on the top half of the case with the number side up. The letters A–L will be on the bottom. Answer the questions in your activity book, close the case and flip it over, and check to see if your answers match. VersaTiles activities can help you learn about numbers and problem solving as you practice and strengthen your math skills.

Mathematics for SylvanSync™
Grades K–2
VersaTiles: Exploring Data

Activity

- Go to page 13 in *VersaTiles: Exploring Data* to practice interpreting and comparing data using a real object graph. Flip over the box to see if you did it correctly.
- Go to page 15 in *VersaTiles: Exploring Data* to practice interpreting and comparing data in a pictograph. Flip over the box to see if you did it correctly.
- Go to page 19 in *VersaTiles: Exploring Data* to practice interpreting and comparing data in a bar graph. Flip over the box to see if you did it correctly.
- Go to page 31 in *VersaTiles: Exploring Data* to practice identifying and listing possible combinations. Flip over the box to see if you did it correctly.

Mathematics for SylvanSync™
Grades K–2
The Problem-Solving Process

Objectives

The student will be able to:

- identify the four basic steps to the problem-solving process.
- use the problem-solving process to solve word problems.

Materials

- none

Background Information

Word problems can be solved using four basic steps. As each step is correctly completed, the solution is one step closer. Look at the following problem. Review the four steps as you find the solution.

Raja planted 9 sunflower seeds. A few weeks later only 5 seeds had sprouted. How many seeds did not sprout?

Step 1: **UNDERSTAND** what is known. What question is being asked?

- Raja planted 9 seeds and 5 seeds sprouted.
- How many did not sprout?

Step 2: Make a **PLAN** to solve the problem.

- Subtraction can be used to solve this problem.

Step 3: **SOLVE** using the plan, show your work. State the final answer in a complete sentence.

- $9 - 5 = 4$; There were 4 seeds that did not sprout.

Step 4: **LOOK BACK** and check the answer. Does it make sense?

- Draw 9 seeds and cross out 5 of the seeds to check.
- How many are left? Four seeds are left.

Mathematics for SylvanSync™
Grades K–2
The Problem-Solving Process

Activity

Use the four basic steps to solve each problem. Record your answers on a separate piece of paper.

Part 1

On a sheet of paper, write the words or numbers to fill in the blanks for each step.

Julia's gymnastics class has 7 students. Mikel's gymnastics class has 9 students. How many gymnastics students are there in all?

Step 1: UNDERSTAND what is known and what is being asked.

There are _____ students in one class and _____ students in another. How many _____?

Step 2: Make a PLAN to solve the problem.

_____ the number of students in each class.

Step 3: SOLVE using the plan.

$7 + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$; There are _____ students in all.

Step 4: LOOK BACK and check the answer. Does it make sense?

Draw a picture to show your answer.

Mathematics for SylvanSync™
Grades K–2
The Problem-Solving Process

Part 2

Use the information to solve each problem.

1. It was sunny 3 days last week and 6 days this week. How many days has it been sunny in the last two weeks?
 - a. What information is known? What is being asked?
 - b. What plan can be used to solve this problem?
 - c. Solve the problem.
 - d. How can the work be checked? Check the answer.
2. Dagney has 13 guppies in her aquarium. Her neighbor gave her 11 more guppies. How many guppies does Dagney have now?
 - a. What is the first step to solving this problem?
 - b. What plan can be used?
 - c. Solve the problem.
 - d. How can the work be checked? Check the answer.

Mathematics for SylvanSync™
Grades K–2
The Problem-Solving Process

Part 3

Use the four-step process to solve each problem. Show all work.
Answer each question in a complete sentence.

1. Darrell has 3 blocks. Joyce has 5 blocks. How many blocks do they have in all?
2. There are 9 birds on the fence. There are 3 birds in the tree. How many birds are there all together?
3. Daman has 4 crayons. Juan has 6 crayons. Sandra has 3 crayons. How many crayons do they have all together?
4. Gretchen spent 5 dollars on a hat and 4 dollars on gloves. How much money did Gretchen spend?

Mathematics for SylvanSync™

Grades K–2

Problem-Solving Strategies

Objectives

The student will be able to:

- choose an appropriate problem-solving strategy.
- use problem-solving strategies to solve word problems.

Materials

- crayons or markers

Background Information

There are different ways to solve word problems. Sometimes a picture or model can be drawn to help answer the question. Using a table or a list can solve some word problems. These different ways of solving word problems are called **strategies**.

Use the four basic steps when solving a word problem.

Louie, Katy, and Philip sit in a row of three. Katy sits in the middle, and Philip sits in the last seat. Where does Louie sit?

Step 1: **UNDERSTAND** what you know, and what is being asked.

Louie, Katy, and Philip sit in a row. Katy sits in the middle and Philip sits in the last seat.

Where does Louie sit?

Step 2: **PLAN** what strategy to use.

Draw a picture of the three seats to solve the problem. Use what is known to put their names on the seats in the row.

First	Middle	Last
		
?	Katy	Philip

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Grades K–2
Problem-Solving Strategies

Step 3: SOLVE the problem.

Louie sits in the first seat in the row.

Step 4: LOOK BACK and check the answer. Does it make sense?

Reread the problem while using the drawing to check if the answer makes sense.

Activity

Solve each problem according to the directions. Record your answers on a separate piece of paper.

Part 1

Using crayons or markers, draw a picture or a model to solve each problem.

1. Alisha drew a red triangle, a blue circle, a yellow square, a red triangle, and a blue circle. She wants to draw two more shapes to complete her pattern. What should she draw?
2. Jackson painted 4 desks, each a different color. He used red, yellow, blue, and orange paint. He put the orange and blue desks next to each other. He did not put the yellow and red desks next to each other. He did not put the red desk next to the orange desk. What was the order of the painted desks?
3. Tammy made 23 cupcakes. She gave 8 to her neighbor and 6 to her grandmother. Then she made 12 more. How many cupcakes does Tammy have?
4. Maggie's dad has five coins in his pocket. The value of the coins is 26 cents. What coins are in his pocket?
5. Justus had 24 cookies to divide evenly among 6 friends. How many cookies did each friend get?

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Grades K–2
Problem-Solving Strategies

Part 2

On your paper, draw tables like the ones shown below for each problem. Solve the problems. Show your work.

1. Jill and Mark run during the week to get in shape for baseball. Jill runs 3 miles every other day. Mark runs 2 miles every day. At the end of day 7, who has run the most miles?

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Jill	3						
Mark	2						

2. Erika has a blue shirt, an orange shirt, a green pair of shorts, and a yellow pair of shorts. She has a black hat and a purple hat. How many different outfits can she make using these clothes? (hint: list each outfit)

Shirt	Shorts	Hat

Mathematics for SylvanSync™**Grades K–2****Problem-Solving Strategies**

3. Jordan's dad offered to make him baked potatoes or mashed potatoes and either pork or chicken. From how many different combinations of potatoes and meat can Jordan select?

Part 3

Work each problem backwards in order to solve it.

1. Jane has 2 more pieces of gum than Jacob does. Jacob has 3 more pieces of gum than Rose does. Rose has 7 pieces of gum. How many pieces of gum does Jane have?
2. Lisa left 5 minutes before Cassie to get to the game. Cassie left 30 minutes after Ashley. Ashley left at 1:00. What time did Lisa leave to get to the game?
3. Misty went shopping for gifts for her sister and brothers. She spent \$5.00 on Lou, \$4.00 on Jay, and \$3.00 on Keyana. She bought herself a juice box for \$1.00. She returned home with \$7.00. How much money did Misty have before going shopping?

Mathematics for SylvanSync™

Grades K–2

Flash Cards

Objectives

The student will be able to:

- demonstrate mastery of basic addition facts.
- demonstrate mastery of basic subtraction facts.
- demonstrate mastery of both basic addition and subtraction facts.

Materials

- addition and subtraction flash cards
- addition and subtraction triangle flash cards
- color tiles, optional

Background Information

Addition means combining groups or numbers. The answer to an addition problem is the **sum**.

Find the sum of the groups of triangles.

$$\begin{array}{ccccccc} \triangle\triangle\triangle\triangle & + & \triangle\triangle & = & \triangle\triangle\triangle\triangle\triangle\triangle \\ 5 & + & 3 & = & 8 \end{array}$$

Subtraction means taking one group or number away from another group or number. The answer to a subtraction problem is the **difference**.

Four flowers are taken away from a group of ten flowers. What is the difference?

$$10 - 4 = 6$$

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Grades K–2

Flash Cards

Activity

Use the flash cards to practice addition and subtraction. If desired, model the number sentences using color tiles. Record your answers on a separate piece of paper.

Part 1

Shuffle the deck of the addition flash cards. Pick a card. Find the sum. Check the answer on the other side of the card.

Write the addition number sentence and the answer. Repeat until you have answered 20 cards correctly.

Part 2

Shuffle the deck of the subtraction flash cards. Pick a card. Find the difference. Check the answer on the back of the card.

Write the subtraction number sentence and the answer. Repeat until you have answered 20 cards correctly.

Part 3

Shuffle the deck of triangle flash cards. Arrange the cards so the purple numbers are at the top. Draw a card. Use a small piece of paper or your hand to cover up the purple number.

Write two addition number sentences using the two blue numbers. Find the sums. Uncover the purple number to check your answers. Repeat until you have answered 20 cards correctly.

Part 4

Shuffle the deck of triangle flash cards. Draw a card. Use a small piece of paper or your hand to cover up one of the blue numbers.

Write the subtraction number sentence using the purple and blue numbers. Find the difference. Uncover the other number to check your answer. Repeat until you have answered 20 cards correctly.

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Grades K–2
Flash Cards

Part 5

Draw five triangles on your paper. Place a subtraction sign in the middle of two of the sides and an addition sign in the middle of the third side.

Shuffle the deck of triangle flash cards. Draw a card. Look at the three numbers on the card. Insert the numbers in the correct corners in the triangle on your paper.

Repeat until your triangles are completed.

Part 6: *Optional*

Shuffle the triangle flash cards. Draw a card. On your paper write the addition and subtraction facts related to the three numbers.

Repeat ten times with ten different cards.

Mathematics for SylvanSync™

Grades K–2

Counting

Objectives

The student will be able to:

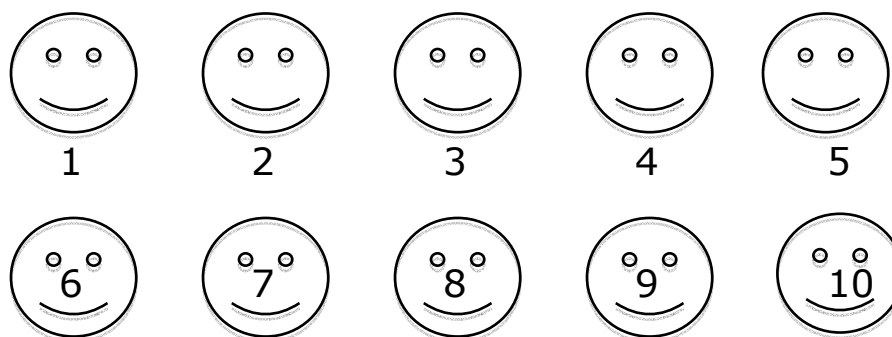
- count 1 to 10 using objects.

Materials

- bear counters
- dice
- crayons, optional
- copy of Worksheet

Background Information

Look at the faces below. How many faces are there? Count by touching each face.



There are 10 faces.

Count the A's and circle the correct number of A's.

A A A A A A A

4 7 9

Mathematics for SylvanSync™
Grades K–2
Counting

Activity

Record your answers on a separate piece of paper.

Part 1

Place a bear on each box and count the number of bears. Write the number of bears.

1.

--	--	--

2.

--	--	--	--	--

3.

Mathematics for SylvanSync™
Grades K–2
Counting

Part 2

Count the hearts. Put a bear below each heart. Count the total number of hearts and bears. Write the total number of hearts and bears.

1.



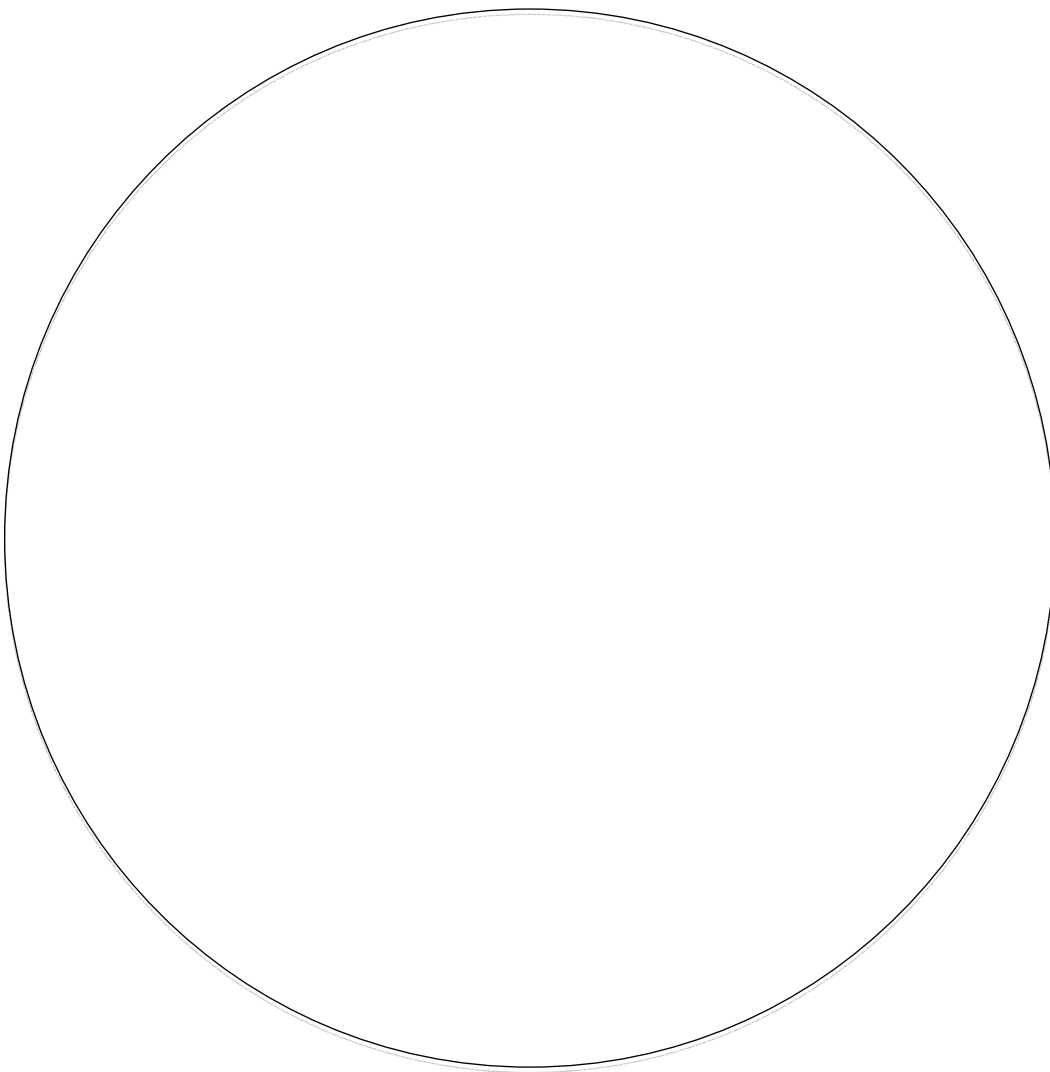
2.



Mathematics for SylvanSync™
Grades K–2
Counting

Part 3

Roll the dice. Count the same number of bears and place inside the circle. Write the number of bears. Repeat three times.

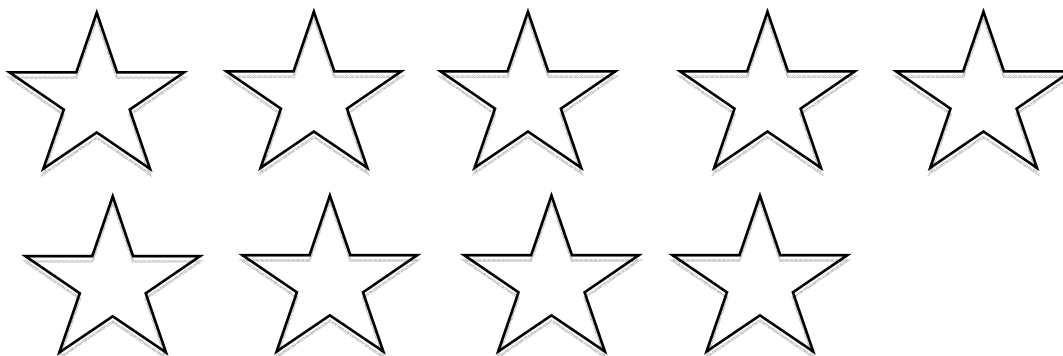


Mathematics for SylvanSync™
Grades K–2
Counting

Part 4 Worksheet

For each problem, count the stars. Then draw and color the same number of squares on your paper to show how many stars there are.

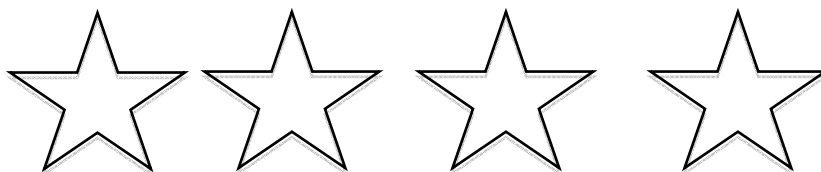
1.



2.



3.



Mathematics for SylvanSync™

Grades K–2

Sorting

Objectives

The student will be able to:

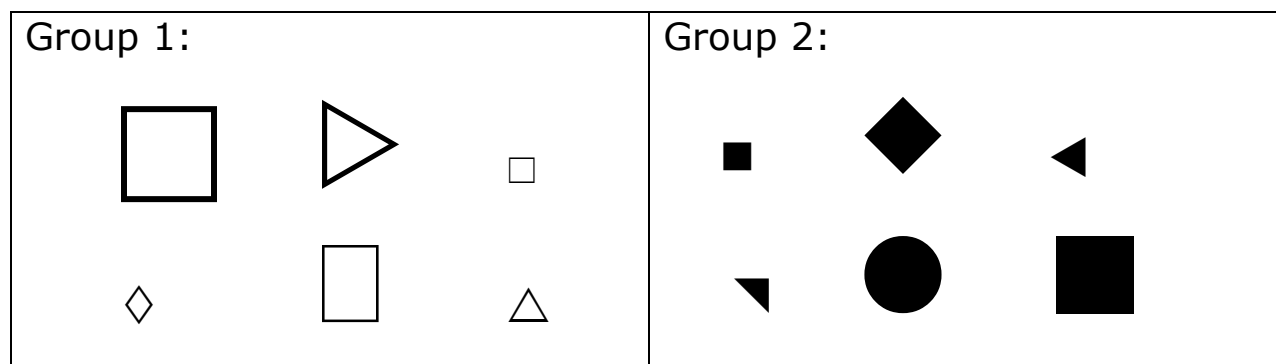
- sort objects based on attributes.

Materials

- bear counters
- attribute blocks
- markers or crayons

Background Information

Look at the shapes below. They are sorted according to their color. The white shapes are one group, and the black shapes are another group.

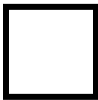


Mathematics for SylvanSync™

Grades K–2

Sorting

Another way to sort the shapes is by their size. The larger shapes are one group and the smaller shapes are another group.

Group 1:      	Group 2:      
---	---

Activity

Record your answers on a separate piece of paper.

Part 1

Use the bear counters and sort as follows.

1. Sort the bears according to size.
2. Sort the bears according to color.

Part 2

Use the attribute blocks and sort as instructed.

1. Sort the blocks according to shape.
2. Sort the blocks according to color.
3. Sort the blocks according to the number of sides on each block.

Mathematics for SylvanSync™
Grades K–2
Sorting

Part 3

Select any bear. Use the color and size to complete the sentence below.

1. This bear is _____ and _____.
2. Find three more bears that can be described the same way.
Draw and color a picture of the group of bears.

Part 4

Using the bear counters and the attribute blocks, draw and color a group according to the directions.

1. Make a group of five bears that are the same color.
2. Make a group of six bears that are the same size.
3. Make a group of eight bears that are the same size and the same color.
4. Make a group of ten blocks that are the same shape.
5. Make a group of nine blocks that are the same size.
6. Make a group of four blocks that have the same number of sides.

Mathematics for SylvanSync™

Grades K–2

Pascal's Triangle

Objectives

The student will be able to:

- identify odd and even numbers.

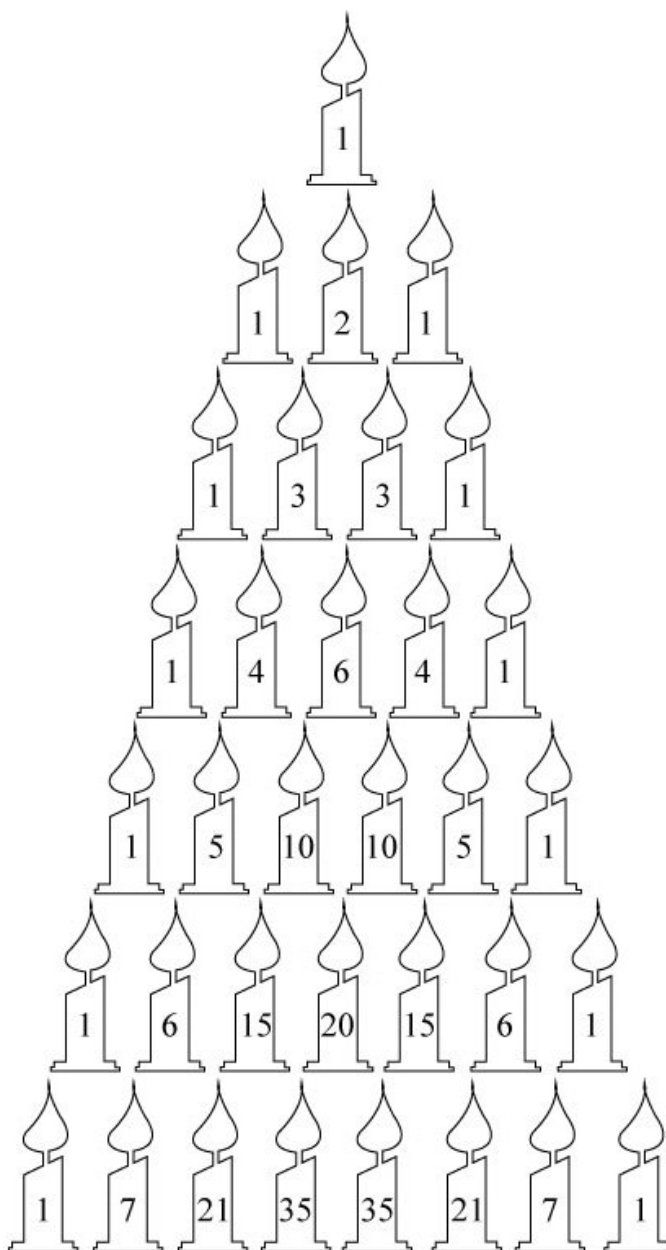
Materials

- colored pencils or crayons
- copy of Worksheet 1
- copy of Worksheet 2

Background Information

Look at the candles on the next page. Look for patterns.

Mathematics for SylvanSync™
Grades K–2
Pascal's Triangle



This arrangement of numbers is called **Pascal's Triangle**. It reveals many important and interesting patterns.

A few discoveries can be made about odd and even numbers using Pascal's Triangle.

Mathematics for SylvanSync™
Grades K–2
Pascal's Triangle

Activity

Use Worksheets 1 and 2 to find patterns in odd and even numbers and their sums. Record your answers on a separate sheet of paper.

Part 1

Using Worksheet 1, color the odd numbers yellow and the even numbers purple. Then continue with the exercises.

1. The color of the right and left sides of the triangle is _____. This is because the number 1 is an _____ number.
2. Find two yellow candles that are side by side. Look at the candle below them. Repeat with two other yellow candles. Repeat again. What is the pattern?
3. Find two purple candles that are side by side. Look at the candle below them. Repeat with two other purple candles. Repeat again. What is the pattern?
4. Look for a pattern when a yellow candle is next to a purple candle. What is true about the candle below them?

Mathematics for SylvanSync™
Grades K–2
Pascal's Triangle

Part 2

Use Worksheet 2 to complete the exercises.

1. Is there a pattern of repeating numbers within each row? Explain.
2. Place a finger on a number. Look at the two numbers above it. What is true about these numbers?
3. Place two fingers on two numbers side by side. What is true about the number below them?
4. Use this pattern to complete the last two rows in the triangle.
5. Color the even numbers and the odd numbers in two different colors.

Part 3

Complete the sentences below.

1. The sum of two odd numbers is an _____ number.
2. The sum of two even numbers is an _____ number.
3. The sum of an even number and an odd number is an _____ number.

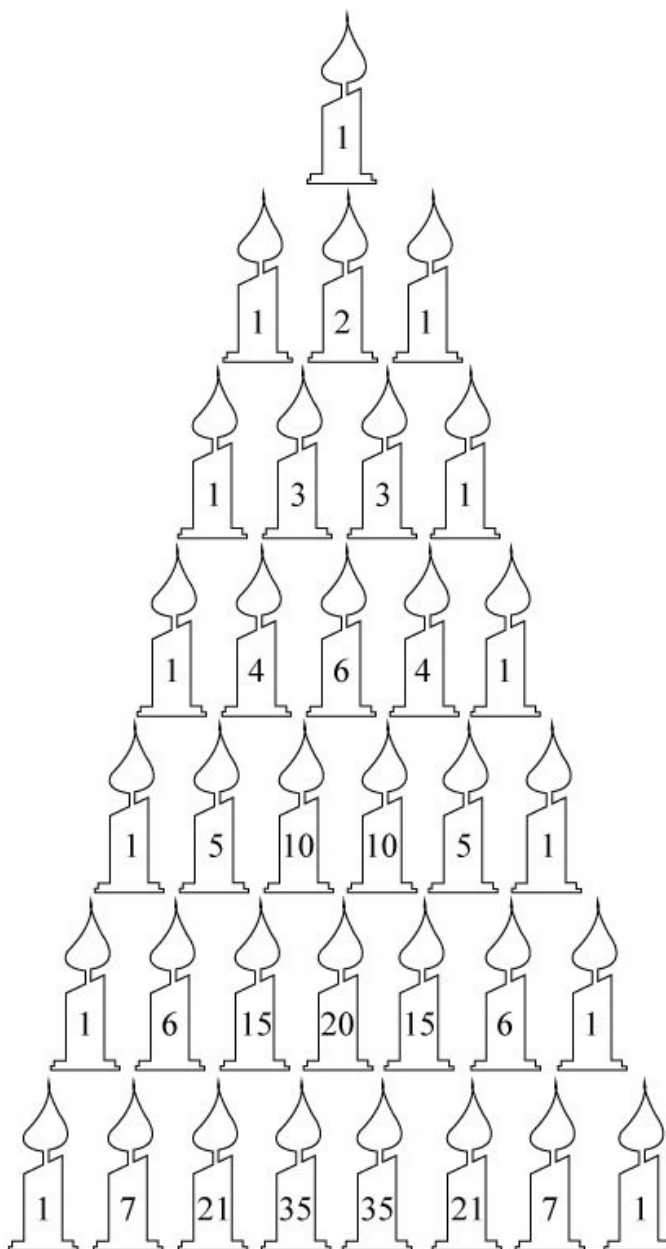
Part 4

Continue to use Worksheet 2 to complete the exercises.

1. Fold the triangle in half lengthwise. What is true about the colors?
2. Describe any other patterns you see.

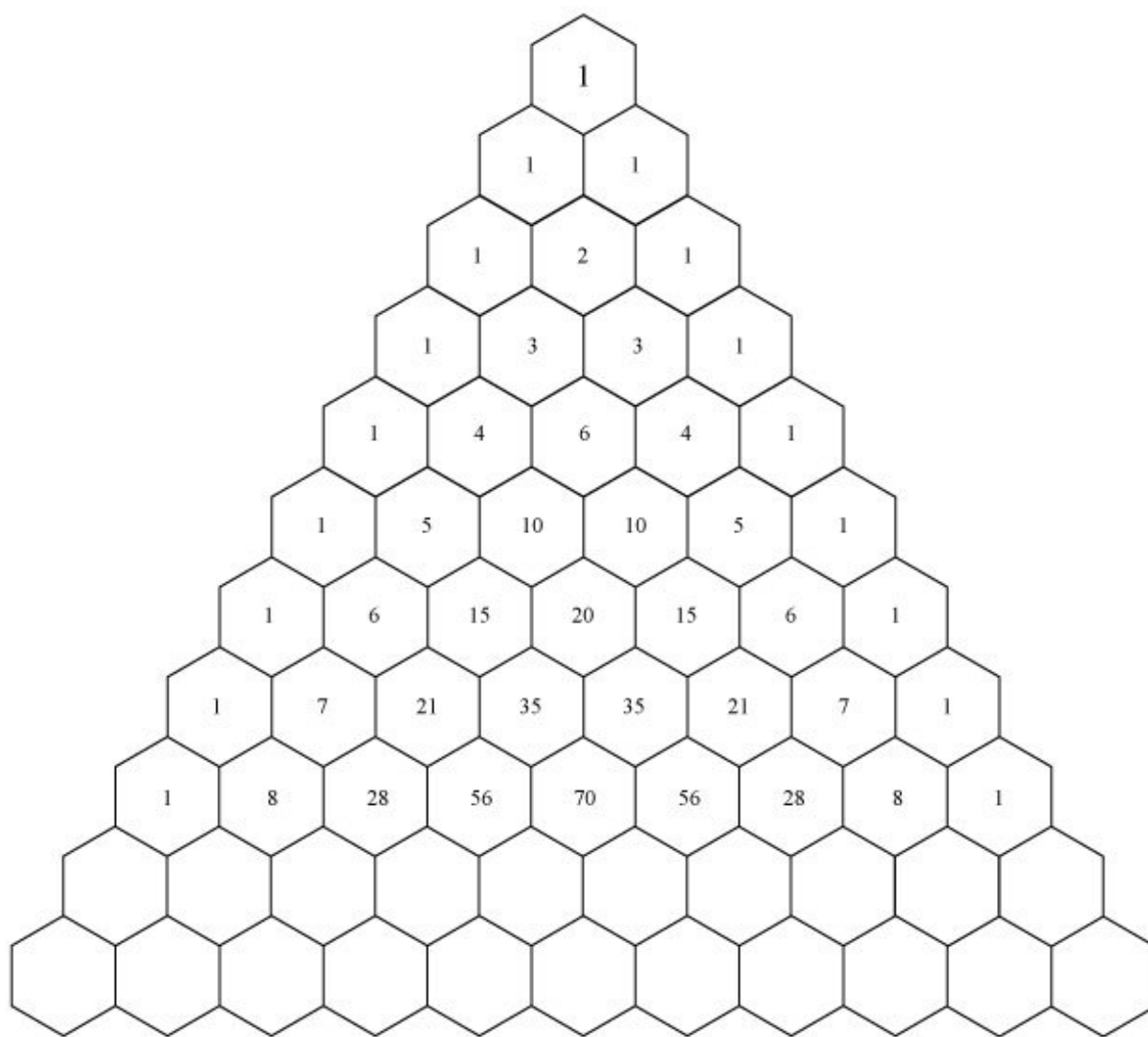
Mathematics for SylvanSync™
Grades K–2
Pascal's Triangle

WORKSHEET 1



Mathematics for SylvanSync™
Grades K–2
Pascal's Triangle

WORKSHEET 2



Mathematics for SylvanSync™

Grades K–2

Comparing Sets

Objectives

The student will be able to:

- identify more or less.
- identify equal sets.

Materials

- crayons or markers
- bear counters
- two-color counters

Background Information

	5 hands
	fewer than 5 hands
	same number of hands
	more than 5 hands

Mathematics for SylvanSync™
Grades K–2
Comparing Sets

Activity

Complete the activities that follow on a separate sheet of paper.

Part 1

Make a table like the one below on your paper.

1. Place 9 counters in the top row.
2. Place more than 9 counters in the second row.
3. Place fewer than 9 counters in the third row.
4. Draw each counter on your paper as you remove them.

Mathematics for SylvanSync™
Grades K–2
Comparing Sets

Part 2

Make a table like the one below on your paper for each of the following activities.

1. Draw 5 circles in the top row.
Draw 9 circles in the second row.
Circle the group that has more.
2. Draw 7 stars in the top row.
Draw 5 stars in the second row.
Circle the group that has more.
3. Draw 8 squares in the top row.
Draw 9 squares in the second row.
Circle the group that has more.
4. Draw 8 diamonds in the top row.
Draw 11 diamonds in the second row.
Circle the group that has fewer.
5. Draw 11 hearts in the top row.
Draw 8 hearts in the second row.
Circle the group that has fewer.

Mathematics for SylvanSync™
Grades K–2
Comparing Sets

Part 3

Use bear counters to complete the activities. Draw and color a picture to show the result for each activity.

1. Select a handful of bear counters.
2. Create a group of fewer bears.
3. Create a group of more bears.
4. Create a group of the same number of bears.

Mathematics for SylvanSync™
Grades K–2
Introducing Base 10

Objectives

The student will be able to:

- represent numbers using base 10 blocks.
- identify a number represented by base 10 blocks.

Materials

- base 10 blocks
- number cubes

Background Information

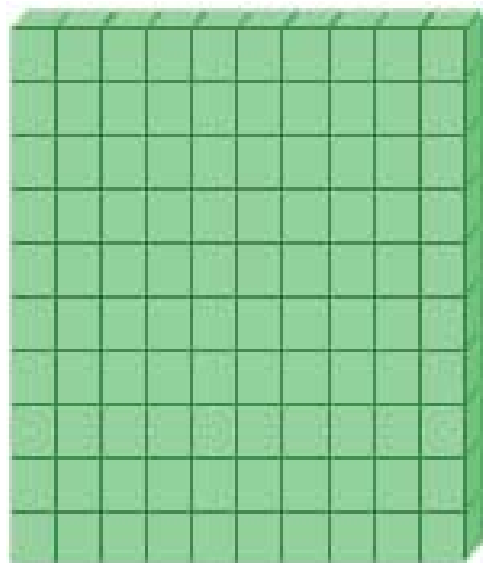
Base 10 blocks can be used to model numbers.



= ones



= tens



= hundreds

Mathematics for SylvanSync™
Grades K–2
Introducing Base 10

What number is modeled below?



6 tens + 8 ones

68

Activity

Record your answers on a separate sheet of paper.

Part 1

Roll two number cubes. Write the number as a two-digit number. Model the number using base 10 blocks. Repeat ten times. Draw pictures of your answers.

Part 2

Roll three number cubes. Write the number as a three-digit number. Model the number using base 10 blocks. Repeat ten times. Sketch your answers.

Mathematics for SylvanSync™

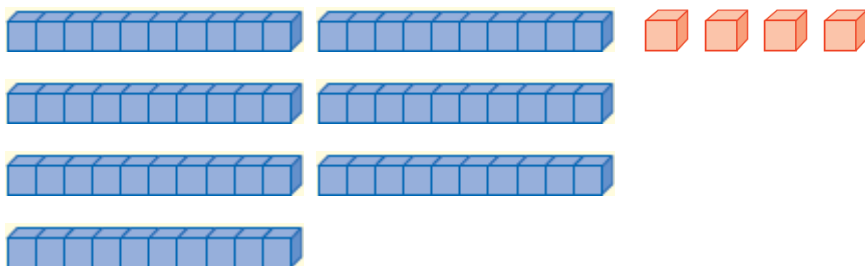
Grades K–2

Introducing Base 10

Part 3

Identify each number represented below.

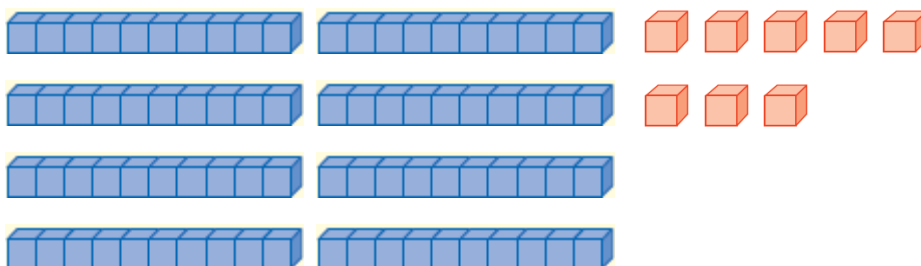
1.



2.

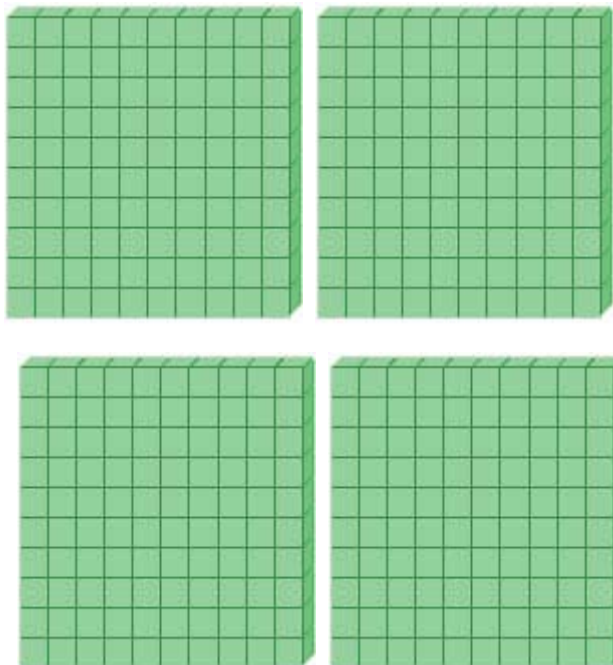


3.

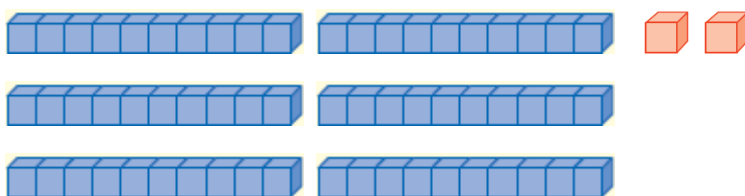


Mathematics for SylvanSync™
Grades K–2
Introducing Base 10

4.

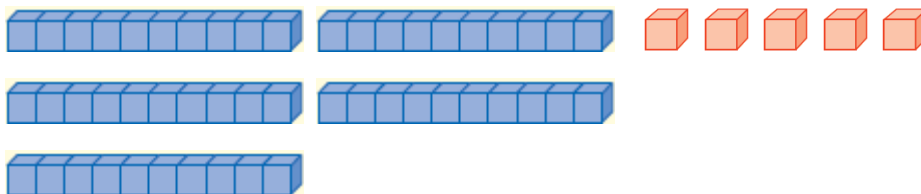


5.

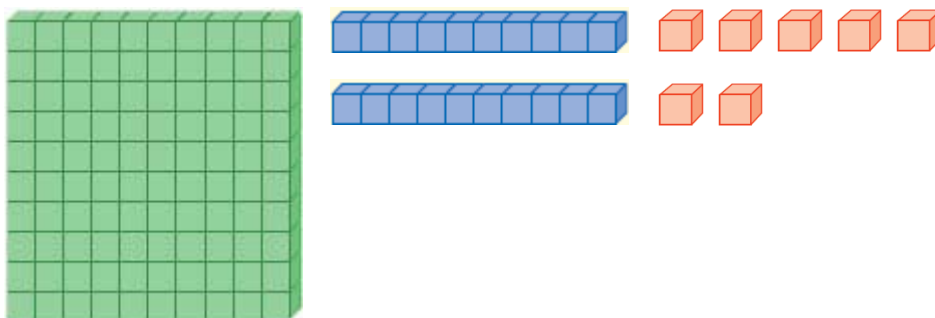


Mathematics for SylvanSync™
Grades K–2
Introducing Base 10

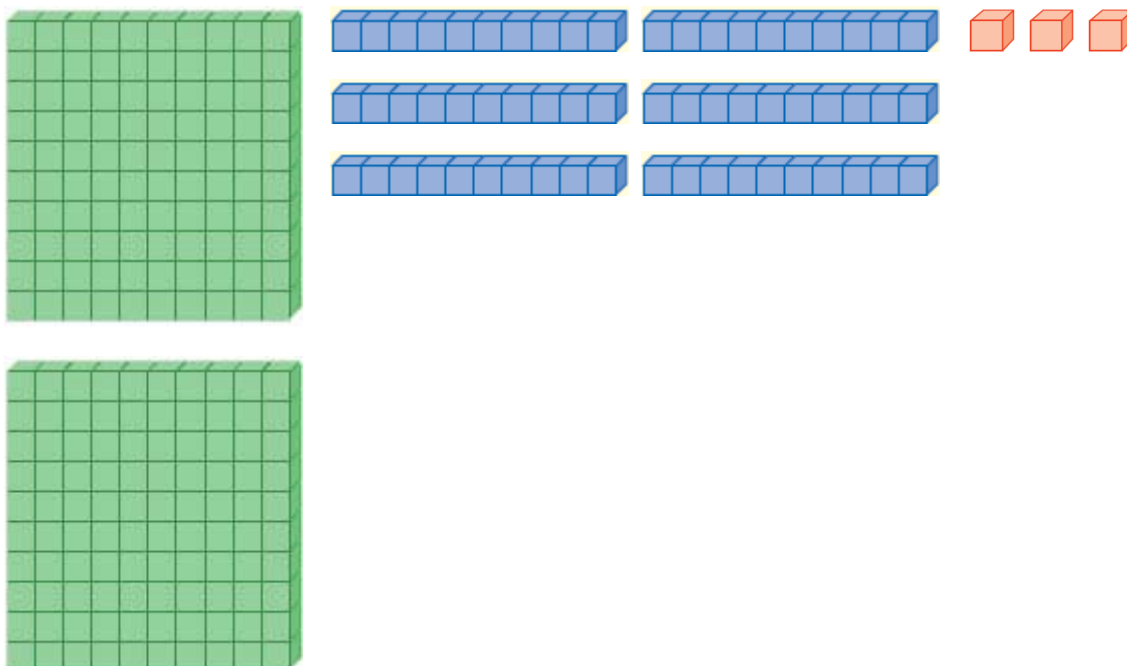
6.



7.



8.



Mathematics for SylvanSync™

Grades K–2

Shapes in the Plane

Objectives

The student will be able to:

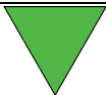
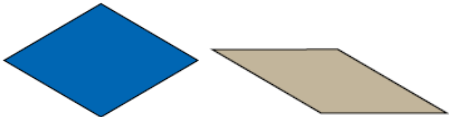
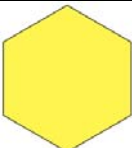
- identify geometric shapes.

Materials

- pattern blocks
- colored pencils

Background Information

Review the shapes of your pattern blocks.

Name	Picture	Number of sides
triangle		3
square		4
trapezoid		4
rhombus		4
hexagon		6

Mathematics for SylvanSync™
Grades K–2
Shapes in the Plane

Activity

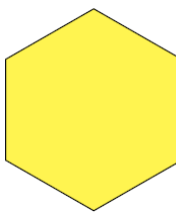
Record your answers on a separate sheet of paper.

Part 1

Match each name with the correct shape. Some will have more than one answer.

1. Trapezoid

a.



2. Rhombus

b.



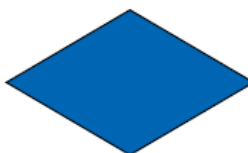
3. Square

c.



4. Triangle

d.



5. Hexagon

e.



Mathematics for SylvanSync™
Grades K–2
Shapes in the Plane

Part 2

Use pattern blocks to complete the exercises.

1. Trace one of each different shape on your paper. Color the shape and write the name next to the shape. Then describe each shape in as much detail as you can.
2. Copy the table. Sort your shapes and sketch each in the appropriate category. In the second row, sketch other examples of shapes that would be in the same categories but are not part of your pattern block set.

3 sides	4 sides	6 sides

3. Sort the different 4-sided shapes into different categories. Explain how you grouped the shapes.
4. Look around the room. Describe any of these shapes that you see.

Mathematics for SylvanSync™
Grades K–2
Shapes in the Plane

Part 3

Follow the instructions to create more shapes. Sketch and color your results.

1. Use six triangles to create a hexagon.
2. Use a triangle and a trapezoid to create a new triangle.
3. Use two triangles to create a rhombus.
4. Create a square using four squares.
5. Create a trapezoid using three triangles.
6. Use three blue rhombuses to create a hexagon.
7. Use two light brown rhombuses and a blue rhombus to create a hexagon.

Mathematics for SylvanSync™
Grades K–2
Shapes in the Plane

Part 4

Create five new shapes on your own! Describe which shapes you use and sketch each new creation.

Part 5: *Optional*

Put some blocks together and outline the result (only the outer edges!). Remove your blocks. Repeat 5 five times for five new shapes. Now go back to each outline and see if you can put the shape back together again using your pattern blocks.

Mathematics for SylvanSync™

Grades K–2

Graphs and Graphing

Objectives

The student will be able to:

- read a picture graph.
- create a picture graph.

Materials

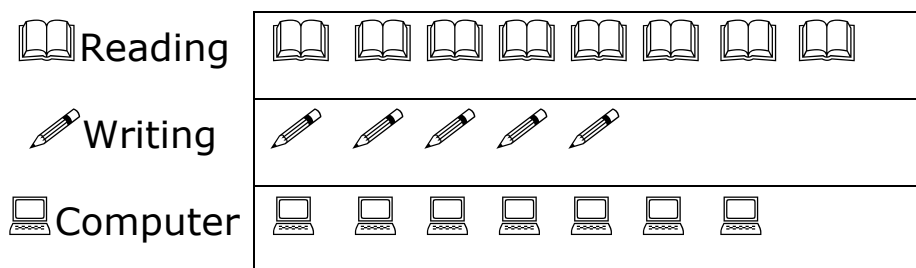
- bear counters
- attribute blocks
- colored pencils or markers

Background Information

A **pictograph** uses pictures to represent information. Pictographs can be used to study and compare the information.

The pictograph below shows the afternoon activities chosen by students. Each symbol shows that one student chose that activity.

Afternoon Activities for Students



Which activity did the most students chose?

Count the number of students who chose each activity.

Eight students chose reading, 5 students chose writing, and 7 students chose computer. The most students chose reading.

Mathematics for SylvanSync™
Grades K–2
Graphs and Graphing

Activity

Use the given information to complete the exercises. Record your answers on a separate piece of paper.

Part 1

The second grade students recorded the number of books they read together in class during four weeks of school. They created a pictograph to study the information.

Week 1	    
Week 2	       
Week 3	  
Week 4	         

Key:  = 1 book

1. During which week did the second grade read the most books?
2. During which week did the second grade read the fewest books?
3. How many books were read during weeks 1 and 2?
4. How many books were read in all?
5. School was cancelled for two days during one week because of snow. Using the pictograph, which week do you think the second grade did not have school? Explain.

Mathematics for SylvanSync™
Grades K–2
Graphs and Graphing

Part 2

Kyle asked his friends to vote for what they wanted to do for his party. He made a chart to keep track of the votes. Kyle did not vote, and each friend only voted once.

Kyle's Birthday Party

Activity	Tally Marks
 Fishing	
 Bowling	
 Skating	

Key: | = 1 vote

1. Which activity got the most votes?
2. How many more friends voted for skating than for fishing?
3. How many friends voted?
4. If Kyle adds his vote for skating, what is true about the number of votes for bowling and skating?

Mathematics for SylvanSync™
Grades K–2
Graphs and Graphing

Part 3

Without using circles, make a group of 20 attribute blocks. Copy the pictograph shown below. Complete the pictograph to show the number of three-sided, four-sided, and six-sided blocks that are in the group. Each symbol you draw should represent one block.

Attribute Blocks Pictograph

Three sides	
Four sides	
Five sides	

Key: 1 symbol = one block

Mathematics for SylvanSync™
Grades K–2
Graphs and Graphing

Part 4

Gather 20 bear counters.

1. Create a pictograph to represent the green, blue, red, and yellow bears in your group. Make a key for the pictograph.
2. If the key shows that each symbol represents 2 bears, how would you represent 3 green bears?
3. Using the 20 bears you gathered, create a new pictograph with the new key where each symbol represents 2 bears.

Mathematics for SylvanSync™

Grades K–2

Hundreds Chart

Objectives

The student will be able to:

- count numbers 1 to 100.
- identify number patterns.

Materials

- copy of Worksheet 1
- copy of Worksheet 2
- penny or base 10 unit block
- crayons or colored pencils

Background Information

A 100 chart can help you count, add, subtract, and play with numbers! It can also be helpful in identifying useful number patterns.

Activity

Use copies of the 100 chart to complete the activities.

Part 1

Using Worksheet 1, point to the following numbers on the chart.

1. your birthday
2. the number of siblings you have
3. how old you are
4. the number of people in the room with you right now
5. the number of cookies you wish you could eat

Mathematics for SylvanSync™
Grades K–2
Hundreds Chart

Part 2

Use Worksheet 1 to answer the questions and complete the exercises.

1. What pattern do you notice in the first column?
2. What pattern do you notice in the second column?
3. Put your finger on 24. What is the number below it? What is the difference between these two numbers? Is this true for any number on the chart and the number below it?
4. Put your finger on any number. Look at the number above it. What is true about these two numbers?

Part 3

Use Worksheet 2 to answer the questions and complete the exercises.

1. Complete the 100s chart by filling in the correct numbers.
2. Color all even numbers red and all odd numbers blue.
3. What is true about all numbers with 5 in the ones place?
4. What is true about all numbers with 0 in the ones place?

Mathematics for SylvanSync™
Grades K–2
Hundreds Chart

Part 4

Use Worksheet 1 or 2 to answer the questions and complete the exercises.

1. Place your penny or unit block on 11. Moving one space at a time to the right, how many moves does it take until you reach 21?
2. Place your penny or unit block back on 11. Can you get to 21 using fewer moves? Describe.
3. When you move one row up or down, what is happening?
4. Place your penny or unit block on 31. Describe how you can move to get to 43.
5. Place your penny or unit block on 67. Describe how you can move to get to 99.

Mathematics for SylvanSync™
Grades K–2
Hundreds Chart

Worksheet 1

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Mathematics for SylvanSync™
Grades K–2
Hundreds Chart

Worksheet 2

		3							
			14	15	16				
21									
		43							50
									60
				75					
				85					
91	92								

Mathematics for SylvanSync™
Grades K–2
Evens and Odds

Objectives

The student will be able to:

- identify an even number.
- identify an odd number.

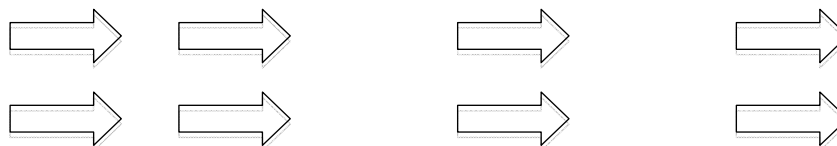
Materials

- bear counters
- cubes

Background Information

An **even** number can be grouped evenly into sets of two. An **odd** number cannot be grouped into sets of two. Odd numbers always have one left over when grouped into pairs.

Look at the eight arrows and circle groups of two with your finger.



Is 8 an even or odd number? The arrows can be grouped into four sets of two evenly, which means eight is an even number.

If the digit in the ones place is even, it is an even number. If the digit in the ones place is odd, it is an odd number.

Mathematics for SylvanSync™

Grades K–2

Evens and Odds

Activity

Record your answers on a separate sheet of paper.

Part 1

Use blocks to represent the given number. Group the blocks into sets of two. Draw your sets. Determine if the number is even or odd. The first one has been done as an example.

	Number	Groups of Two	Even or Odd?
Example:	3		odd
1.	4		
2.	6		
3.	9		
4.	5		
5.	10		
6.	7		
7.	1		
8.	2		
9.	8		

Mathematics for SylvanSync™
Grades K–2
Evens and Odds

Part 2

Use counters to determine if the number is odd or even. Write the correct answer.

1. 14
2. 17
3. 13
4. 20
5. 18

Part 3

Copy each number. Underline the digit in the ones place. Without counters, determine if the number is odd or even. Write the correct answer.

1. 35
2. 29
3. 40
4. 201
5. 136

Part 4

Take a handful of counters. Count them and write the number. Then write odd or even. Repeat four more times.